MIL-S-11581B
8 April 1976
SUPERSEDING
MIL-S-11581A
11 May 1965

MILITARY SPECIFICATION

SCALE, BEAM INDICATING; WEIGHING, SENSITIVE TO 1 GRAM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 20-kilogram (kg) nominal-capacity scale equipped with a 2-kg tare weigh bar.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
AAA-S-121	- Scales, Weighing; General Specifications for.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-F-320	- Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes.

FSC 6670

Source: https://assist.dla.mil -- Downloaded: 202640-17T18:21Z
Check the source to verify that this is the currentversion before use.

MILITARY

MIL-P-116 - Preservation-Packaging, Methods of.

STANDARDS

FEDERAL

FED. STD. No. 356 - Commercial Packaging of Supplies and Equipment.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The scale shall be of the weigh-beam type, with a nominal 20-kg capacity, sensitive to 1 gram, and equipped with a tare weigh bar. The scale shall be in accordance with AAA-S-121 and as specified herein. The maximum overall dimensions shall be as shown in Figure 1.

3.2 Weigh beam. The weigh beam shall be provided with the following:

(a) Main bar, 1000-gram (g) capacity in 100-g increments.
(b) Fractional bar, 100-g capacity in 1-g increments.
(c) Tare weigh bar, 2-kg capacity, between main and fractional bars.
(d) Suspension plate with means of attaching to beam and the following counterpoise weights: one 1 kg, two 2 kg, one 5 kg, and one 10 kg.

3.3 Identification marking. The scales shall be identified in accordance with MIL-STD-130.

3.4 Treatment and painting. Unless otherwise specified (see 6.2), treatment and painting shall be at the option of the contractor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:21Z
Check the source to verify that this is the current version before use.

3.5 Workmanship. All parts, components, and assemblies of the scales including castings, forgings, stampings, and machined surfaces shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous materials.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined as specified in 4.4.1. AQL shall be 1.0 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. The scales shall be examined as specified herein for the following major defects:

Source: https://assist.dla.mil -- Downloaded: 202640-17T18:21Z
Check the source to verify that this is the current version before use.

101. Dimensions not as specified.
102. Components missing or not as specified.
103. Identification or special marking missing or illegible.
104. Workmanship not as specified.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

105. Materials or containers not as specified. Each incorrect material or container shall constitute one defect.
106. Beam and platform not secured as specified for Level A.
107. Cushioning not as specified for Level A.
108. Fiberboard boxes not closed as specified for Level A or B.
109. Strapping not as specified for Level A.
110. Gross weight of wood boxes exceeds 200 pounds for Level A or B.
111. Commercial packaging not in accordance with referenced document.
112. Marking illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Components requiring a preservative in accordance with MIL-P-116 shall be coated with Type P-9 preservative as specified in MIL-P-116. Each scale shall be placed in a fiberboard box conforming to PPP-B-636, W5c. The beam and platform shall be secured to prevent damage and movement by vibration using fiber pads under the platform and under the beam. Cushioning and blocking, using built-up fiber pads, shall be provided to prevent movement or shifting of the scales and components in the container. The fiber pads shall be secured to prevent dislocation during shipment. Fiber pads shall conform to PPP-F-320, Type CF, Class Weather Resistant. The box shall be closed in accordance with Method V of the appendix to the box specification.

Source: https://assist.dla.mil -- Downloaded: 202610-17T18:21Z
Check the source to verify that this is the current version before use.

5.1.2 Commercial. The scales shall be preserved in accordance with FED. STD. No. 356.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A. The scales, preserved as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-621, Class 2, style optional, or PPP-B-601, Overseas Type, style optional. The gross weight of each box shall not exceed 200 pounds. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.2.2 Level B. The scales, preserved as specified in 5.1, shall be packed as specified in 5.2.1 for Level A, except that the boxes shall be Domestic Type. Strapping shall be either Finish A or B. As an alternative, boxes shall conform to PPP-B-640, Class 2. Fiberboard boxes shall be waterproof and closed as specified in the appendix to the box specification.

5.2.3 Commercial. Commercial packing shall be in accordance with the requirements of FED. STD. No. 356.

5.3 Marking.

5.3.1 Levels A and B. Levels A and B marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial marking shall be in accordance with FED. STD. No. 356.

6. NOTES

6.1 Intended use. The scale is intended for weighing commercial samples, animals, compaction studies, mechanical analysis of aggregates, and in making solutions.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When treatment and painting will not be at the option of the contractor (see 3.4).
(c) Degree of preservation and packing required (see 5.1 and 5.2).
(d) When strapping shall be other than as specified (see 5.2.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:21Z
Check the source to verify that this is the current version before use.

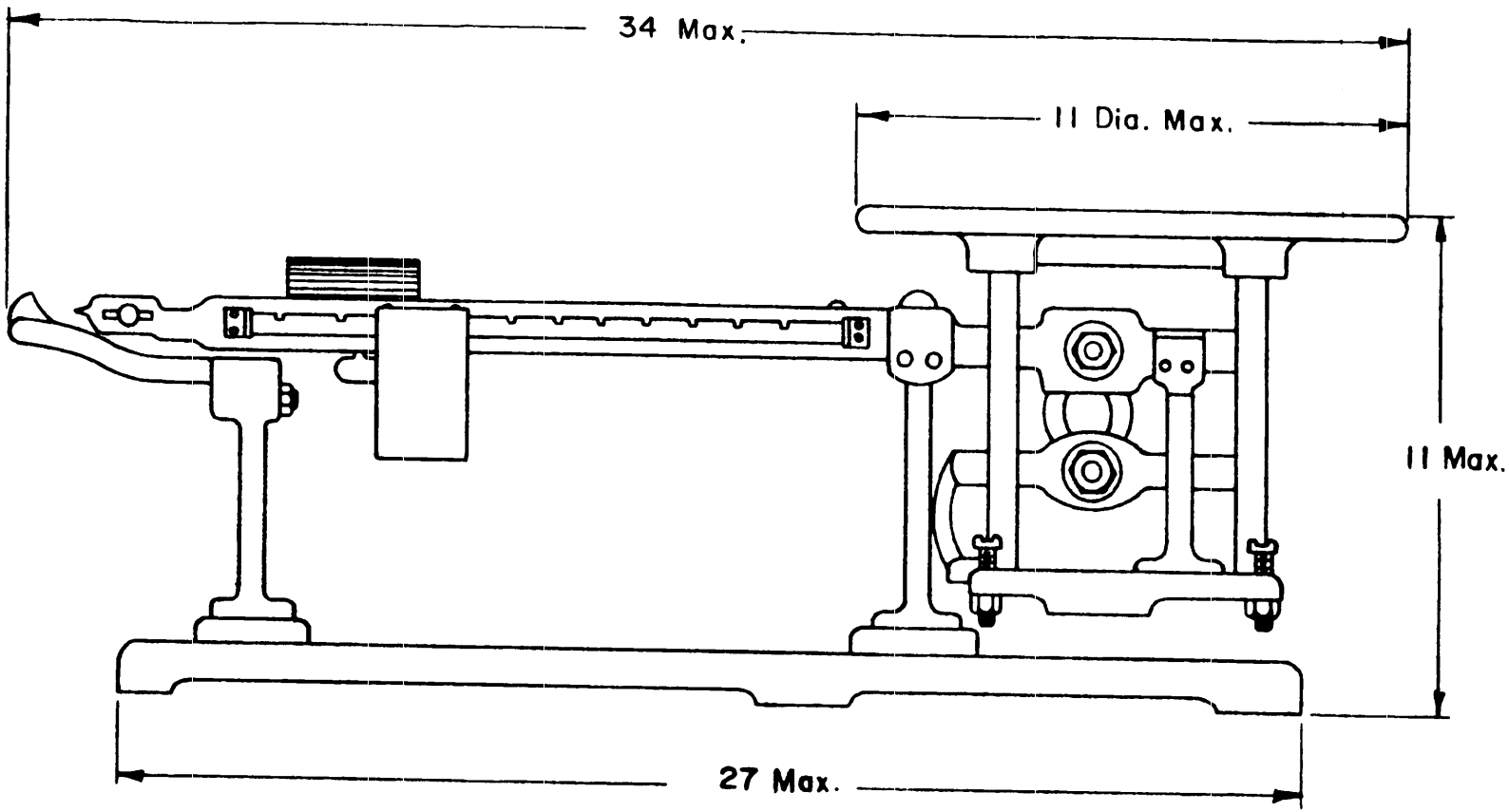
Custodians:
Army - ME
Air Force - 82

Review activities:
Army - GL, SM
DSA - GS

Preparing activity:
Army - ME

Project 6670-0109

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:21Z
Check the source to verify that this is the current version before use.



NOTE:
ALL DIMENSIONS ARE IN INCHES.

FIGURE 1. Scale, beam indicating.

X-177A

Source: https://assist.dla.mil -- Downloaded: 2026-01-17T18:21Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T18:21Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE MIL-S-11501B Scale, Beam Indicating; Weighing, Sensitive to 1 Gram

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 202610-17T18:21Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
US Army Mobility Equipment Research
and Development Command
ATTN: DRXFB-DS
Ft. Belvoir, VA 22060

FOLD

Source: https://assist.dla.mil -- Downloaded: 202610-17T18:21Z
Check the source to verify that this is the current version before use.